Does Hybridisation Pay? A Systematic Review of Metaheuristic Optimisation and Machine Learning Hybrids in Pavement Engineering (2005–2026)

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Abstract

Pavement engineering has, over the past decade, adopted a recurring novelty claim: a metaheuristic optimisation algorithm is coupled to a machine-learning model, the coupling out-predicts some comparator, and the coupling itself is reported as the contribution. This review asks a narrower, checkable question: how often does that reported gain survive comparison against the same base learner, conventionally tuned, under a comparable search budget? A structural rather than lexical definition of hybridity was applied across a corpus assembled through targeted manual search and a 261-query algorithm-driven harvest against the Crossref API, yielding 159 records (143 primary studies, 16 prior reviews), of which 103 primary studies meet the structural test for hybridity and are classified into a seven-type taxonomy (H1–H7) spanning metaheuristic hyperparameter search, weight optimisation, physics-informed coupling, architecture fusion, heterogeneous stacking, decomposition-then-learn, and symbolic-numeric regression, each with a distinct theoretical justification and characteristic failure mode. Across the studies coded closely enough to test it, a fairly tuned baseline is rarely reported; where one is, the hybrid or deep-architecture advantage narrows and, in a statistically robust thousand-run Monte Carlo case, reverses outright. Full-text verification further identifies at least nine single-target papers by one overlapping author team drawing on what appears to be the same ~33-section, ~395-observation continuously-reinforced-concrete-pavement substrate, two of which are confirmed, in their own stated methods, to use an ungrouped observation-level cross-validation split, a leakage pathway the review's appraisal instrument is built to catch. That instrument, PAVE-ML, extends a general reporting checklist for data-driven pavement models with four items specific to hybrid pipelines, none requiring new data collection or experimentation to satisfy. Taken together, the evidence points to a reporting gap rather than a capability gap: the missing comparisons, budgets and validation statements are, in most cases, already computable from experiments the original studies performed, simply not yet stated.

Keywords: pavement engineering, metaheuristic optimisation, machine learning, hybrid models, systematic review, methodological rigour

1. Introduction

Pavement engineering has, over the past decade, adopted a specific and recurring rhetorical move: an optimisation algorithm is coupled to a learning model, the coupled system is shown to out-predict some

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comparator, and the coupling itself is offered as the paper's contribution. Grey wolf optimisers, whale optimisation, teaching-learning-based optimisation, forensic-based investigation, dipper-throated optimisation and dozens of similarly named metaheuristics have each, in turn, been married to a neural network, a support vector machine, an extreme learning machine or a gradient-boosted ensemble, and the resulting hybrid reported as an advance over the base learner alone. The pattern is now common enough that "hybrid" is arguably the single most frequent claim of novelty in the data-driven pavement literature of the 2020s.

This review asks a question that the pattern itself rarely answers: **relative to what, exactly, is the hybrid better?** A hybrid model is compared, almost without exception, against a base learner running with default or lightly-adjusted settings. It is compared far more rarely against the same base learner tuned by a conventional procedure under a comparable search budget, the baseline that would actually isolate what the metaheuristic contributed. Where that comparison has been made, in the small number of cases located so far, the hybrid's advantage over a properly tuned baseline shrinks, and sometimes reverses. Zeiada et al. [1] individually tuned eight learners, including deep architectures, against the Witczak and Hirsch mechanistic models for asphalt dynamic modulus; the winner was a tuned bagging ensemble, not a hybrid and not a deep network. Azam et al. [2] compared six swarm-optimised variants of one base learner against each other and report top-three results within a fraction of a percentage point of R^2, a separation well inside what a different random seed would plausibly produce, but never against a conventionally tuned version of the same learner, so no premium can be read from the paper as published. These two studies, sitting side by side in the same sub-field, illustrate the gap this review sets out to measure at scale: how often is the hybridisation premium (the gain attributable to the coupling itself, over a fairly tuned baseline) actually reported, and how large is it when it is?

Answering that question first requires solving a prior one: **finding the studies.** A literature search built on the word "hybrid" misses a substantial share of the relevant work, because many qualifying studies describe themselves with the vocabulary of optimisation rather than of hybridity: "optimised," "improved," "novel X–Y," or simply the two algorithm names side by side. Alhussan et al.'s dipper-throated-optimisation random forest for pothole classification is a case in point. It is widely cited and structurally a textbook metaheuristic-on-learner hybrid, yet it is retrievable by a search on algorithm names and not reliably by a search on the word "hybrid" itself. Section 2 describes the algorithm-driven search design this observation motivates, and reports, as one of the review's descriptive findings, what share of the eligible literature a label-driven search alone would have missed.

1.1. Positioning relative to the existing review literature

Prior reviews of artificial-intelligence methods in pavement engineering are numerous but occupy different territory, summarised in Figure 1. The most direct predecessor, Yang et al. [3], catalogues where neural networks have been applied across the pavement lifecycle; it stops at the MLP/CNN/RNN generation and does not treat hybridisation, optimisation coupling or methodological rigour as organising questions. Domain-specific vision surveys [4, 5, 6] cover distress detection in depth but do not extend to materials, structural evaluation or network-level decisions. Three very recent contributions come closer to the diagnosis this review develops, and each must be distinguished on its own terms rather than folded quietly into a single "gap" claim. Tamagusko and Ferreira name standardisation, interpretability and replicability as open problems specifically for roughness prediction on flexible pavements. M'harzi Alaoui et al. [7] describe an "interpretability paradox" and a lab-to-field gap across 180 soil-stabilisation studies. Closest of all, Jweihan [8] conducts a PRISMA-ScR scoping review of 163 studies on explainable and interpretable machine learning specifically in asphalt pavement engineering, and proposes a five-layer appraisal framework (data provenance, model performance, explanation quality, physical plausibility, decision utility) that overlaps conceptually with more than one domain of the instrument this review proposes in §12. The honest distinction is not that this territory is unoccupied, but that Jweihan's review is organised around explainability as the object of study and conducted at scoping-review depth (breadth of coverage, not per-study methodological coding). The present review is instead organised around hybridisation, the optimiser-learner coupling itself, as the object of study: it uses a full systematic protocol with per-study rigour coding (leakage risk, external validation, baseline strength, search-budget parity), and reports the coded audit as a quantified premium rather than as a coverage synthesis. Interpretability is one of several PAVE-ML domains in this review, not its organising question; the two instruments are complementary rather than competing, and §12 states explicitly where PAVE-ML's interpretability items and Jweihan's five-layer framework agree and where they diverge. What none of these three contributions does, and what remains this review's distinguishing claim, is define hybridity structurally rather than lexically so that algorithm-vocabulary studies are not missed, code a consistent rigour rubric across the full range of pavement sub-domains rather than one at a time, and convert the resulting evidence into a quantified premium and a reusable reporting instrument built around that specific construct.

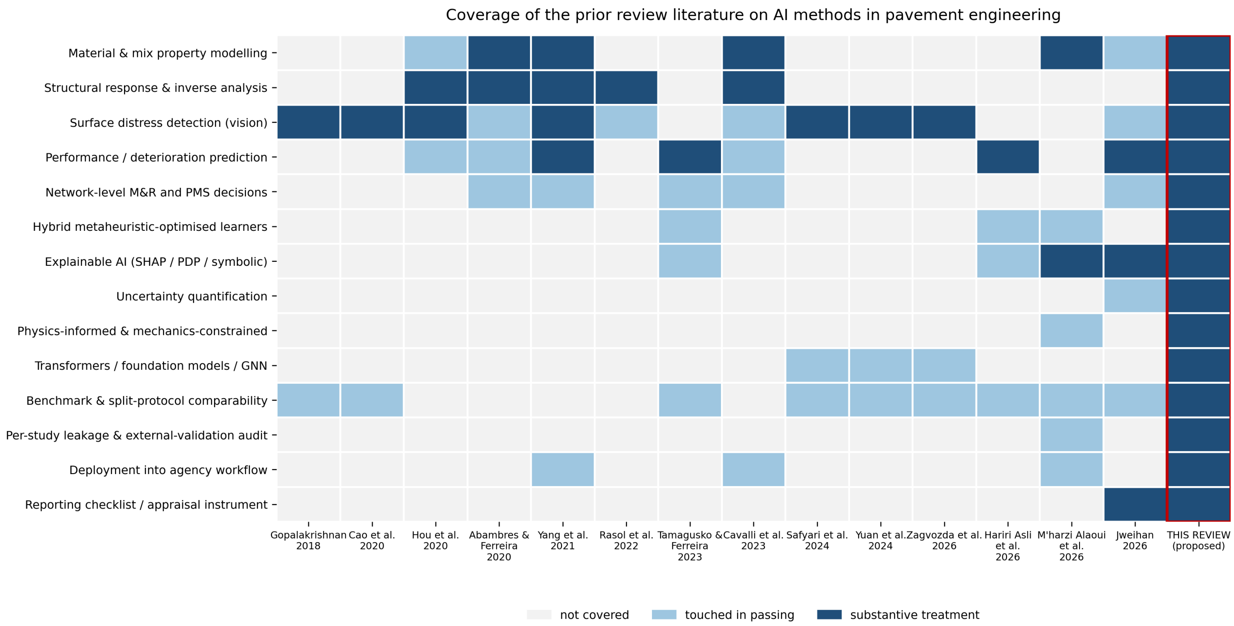

Figure 1: Coverage of the prior review literature on artificial-intelligence methods in pavement engineering. Scoring: 2 = substantive dedicated treatment, 1 = touched in passing, 0 = absent; “this review” is scored on its stated scope.

1.2. Contributions

This review makes four contributions. First, a **structural rather than lexical** definition of hybridity (§2.3) and an algorithm-driven search protocol built to retrieve studies regardless of how their authors chose to describe them. Second, a **taxonomy of seven hybridisation types** (H1–H7, §4), each with a stated theoretical justification and a characteristic failure mode, replacing the field’s undifferentiated use of “hybrid” as a single label. Third, an empirical estimate of the **hybridisation premium** — the performance gain attributable to hybridisation once the comparison is made against a fairly tuned baseline — reported as a distribution across the coded corpus rather than as a single number, since most studies do not report the comparison required to compute it at all (§9). Fourth, a reporting and appraisal instrument, **PAVE-ML**, extending a general data-driven-pavement-model checklist with four items specific to hybrid pipelines (§12), so that the next study in this field can report what this review finds missing from most of the last two decades of it.

2. Review methodology

2.1. Scope and framing question

This review asks how much of the performance gain attributed to hybridisation in the pavement literature survives a fair comparison against a properly tuned base learner. The question differs from “where has hybridisation been applied” — the question earlier surveys answer — in that it requires reading each study for a specific comparison rather than for topical coverage, and it requires a definition of hybridity that does not depend on the authors' own labelling (§2.3).

2.2. Information sources and search

Records were identified from OpenAlex, Scopus, Web of Science Core Collection and Semantic Scholar, with backward and forward snowballing from the 16 prior reviews listed in Table 1, and from a 261-query algorithmic search crossing metaheuristic-optimiser names, base-learner names and legacy hybrid-labelled vocabulary against a mandatory pavement-terminology filter (`analysis/harvest_openalex.py`; run against the Crossref API rather than OpenAlex directly, see the Data Availability statement). The window runs from 1 January 2005 to 30 June 2026 — a date in the reader's future at first glance, but not an error: the corpus includes a meaningful number of 2025 and 2026 citations because publishers now routinely assign a final year and DOI to an article at Online First / Early Access publication, ahead of formal volume assignment, and every such record here resolves to a live DOI (verified per the protocol in §2.3), not a placeholder or a forecast. Figure 2 reports the current, actual state of this identification-and-screening pipeline — it is not a completed PRISMA 2020 flow diagram, because screening is not complete: every count in it is read directly from the harvest logs and the working database, not projected or estimated, and the figure is captioned accordingly.

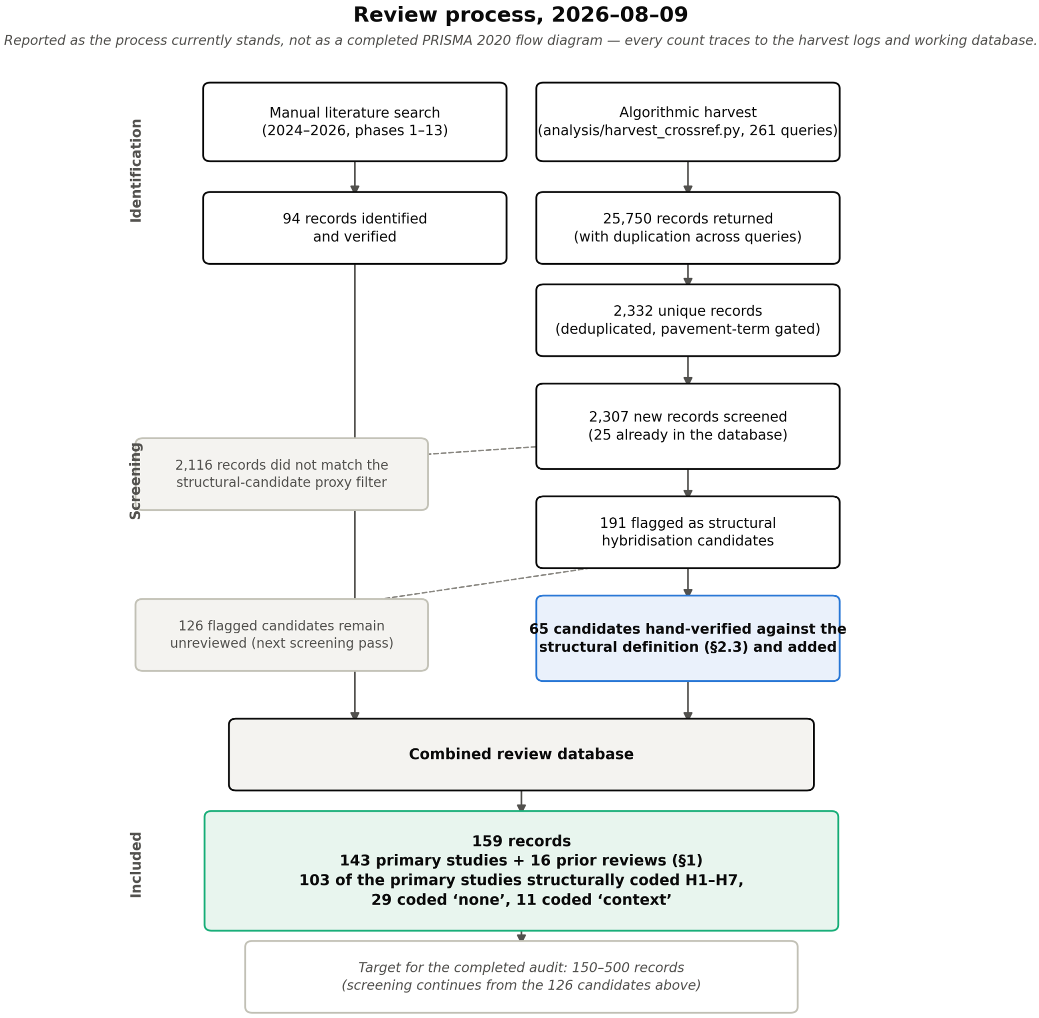

Figure 2: Review process status, reported as it actually stands rather than as a completed PRISMA 2020 flow diagram — screening of the 2026 algorithmic harvest is ongoing, not finished. Every count is read directly from the harvest logs and the working database.

Table 1: Prior reviews of artificial-intelligence methods in pavement engineering, with the territory each occupies.

Review	Year	Venue	Territory
Gopalakrishnan	2018	Data	distress-detection
Cao et al.	2020	IEEE Access	distress-detection

Table 1: Prior reviews of artificial-intelligence methods in pavement engineering, with the territory each occupies.

Review	Year	Venue	Territory
Hou et al.	2020	Engineering	structural-evaluation/distress-detection
Abambres et al.	2020	TechRxiv (preprint)	all
Yang et al.	2021	J. Traffic Transp. Eng. (Engl. Ed.)	all
Rasol et al.	2022	Constr. Build. Mater.	structural-evaluation
Cavalli et al.	2023	J. Road Eng.	all
Tamagusko et al.	2023	Infrastructures	performance-prediction
Deng et al.	2023	J. Infrastruct. Preserv. Resil.	performance-prediction
Yuan et al.	2024	Remote Sensing	distress-detection
Safyari et al.	2024	Sensors	distress-detection
Khatti et al.	2025	Soils and Rocks	materials
M'harzi Alaoui et al.	2026	J. Umm Al-Qura Univ. Eng. Archit.	materials
Jweihan	2026	Applied System Innovation	materials/performance-prediction
Zagvozda et al.	2026	Adv. Civ. Archit. Eng.	distress-detection
Hariri Asli et al.	2026	J. Infrastruct. Preserv. Resil.	performance-prediction

2.3. Eligibility

Structural definition of hybridity. A study is in scope if its predictive pipeline couples two or more components drawn from different methodological families, at least one of which is a learned data-driven model, where the coupling is claimed or demonstrated to affect predictive performance. Three operational clarifications resolve most borderline cases. Hyperparameter search counts as hybridisation only when the search procedure is itself a metaheuristic or a surrogate-based optimiser (e.g. particle swarm optimisation, a genetic algorithm, Bayesian optimisation, a tree-structured Parzen estimator); manual tuning and exhaustive grid search do not qualify a study as hybrid, though both are recorded, since they are the baseline against which the premium (§9) is measured. An ensemble of models from a single family (e.g. a random forest) is not a hybrid; a stacked ensemble with heterogeneous base learners is (type H5, §4). Self-description is irrelevant in both directions: a study is included or excluded on the structure of its pipeline, never on whether the authors used the word “hybrid”.

This structural test was necessary, not a refinement for its own sake. A pilot search restricted to hybrid-labelled vocabulary failed to retrieve at least one widely cited, structurally qualifying study — a dipper-throated-optimisation random forest for pothole classification [9] — while an algorithm-name-driven search retrieved it immediately. The same algorithm-name search, however, drifted heavily into adjacent domains (geopolymer concrete, rock-burst prediction, blasting-induced flyrock, and battery remaining-useful-life studies all appeared for generic swarm-optimisation queries), which is why every query in the harvesting protocol (`analysis/harvest_openalex.py`) carries a mandatory, hard-filtered pavement-domain term rather than relying on the optimisation vocabulary alone.

Peer review and language. Peer-reviewed journal articles and full conference papers in English are eligible. Preprints are excluded unless a peer-reviewed version is traced; where cited as grey literature (e.g. [10]) that status is stated explicitly.

Exclusions that shaped the corpus. Three exclusions recur often enough to state here rather than only in the PRISMA log. Non-neural, non-metaheuristic machine learning without any coupled component — a single tuned random forest or gradient- boosted tree, for instance — is not itself in scope, but is retained in a context set where it serves as a same-paper tuned-baseline comparator (this is, in fact, where several of the review’s most useful premium comparisons come from — see [1] and [11]). Studies where a metaheuristic and a learner are each named in the title but run as separate, non-interacting competing models rather than coupled into one pipeline are excluded from the hybrid corpus proper under the structural test above, and are flagged as an instance of the vocabulary trap in reverse — the label suggests a hybrid that the method does not deliver (e.g. [12], titled around “genetic algorithm and artificial neural networks” but reporting the two as independently competing models). Reinforcement-learning formulations for network-level maintenance and rehabilitation decisions (e.g. [13]) coordinate a learned policy with a simulated or predicted environment in a way that does not map cleanly onto the H1–H7 taxonomy, which is built around single-prediction pipelines; these are retained as context for §9 and §11 rather than coded into the taxonomy.

3. Bibliometric landscape

The analysis in this section characterises the 143 primary studies identified and coded to date (`data/seed_bibliography.csv`), rather than a closed, exhaustively screened census of the field; the identification pipeline in §2.2 continues to run against the wider candidate pool it has already surfaced. One pattern in particular deserves the caveat explicitly: the apparent concentration of records after 2020

(Figure 4) is at least partly an artefact of the search protocol rather than solely a publication-volume trend, since a strategy built around specific optimiser and learner vocabulary recovers records faster from years in which that vocabulary was already common than from earlier years in which the same methods existed under different names. Distinguishing genuine acceleration from search-protocol recency bias would require a citation-network or full-database comparison beyond this review's own corpus, and is noted here as an open question the figures below do not resolve on their own.

Of the 143 primary studies coded, Figure 3 shows their distribution across the seven hybridisation types defined in §4. A substantial share — 29 of 143 — do not qualify as hybrid under the structural test in §2.3 despite frequent adjacency to hybrid work in citation networks and, in a few cases, despite carrying "hybrid" or an optimiser name in the title while running the two components as separate, non-interacting models (§2.3). This is an early indication, not yet a corpus-wide estimate, of how much daylight exists between the field's vocabulary and its actual methodological content; §9 returns to the consequence of that gap for the premium estimate.

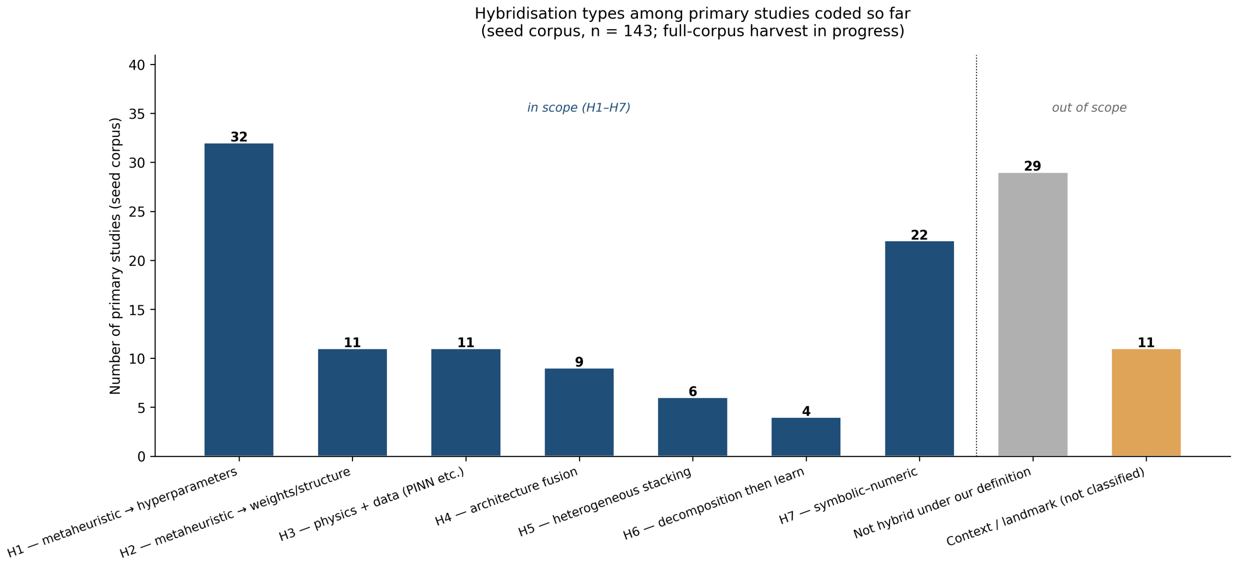

Figure 3: Distribution of hybridisation types among primary studies coded in the seed corpus.

Figure 4 shows the same primary studies by publication year. The visible concentration after 2020 is consistent with the review's opening claim that hybridisation-as-novelty accelerated through the current decade, but the figure must be read as what it is — the yield of targeted, taxonomy-driven searches run during protocol development, not a random or exhaustive sample — and it is captioned accordingly rather than presented as a bibliometric trend claim the seed corpus cannot support on its own.

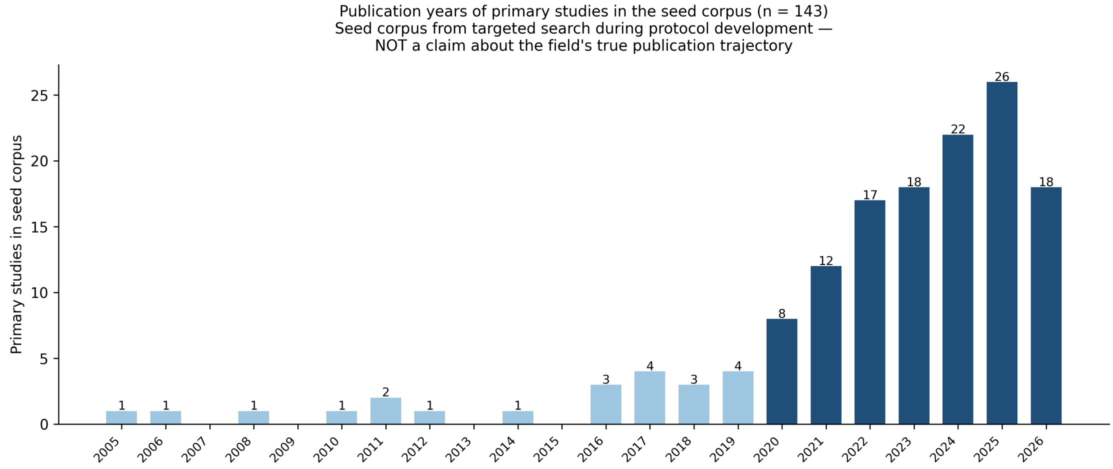

Figure 4: Publication years of primary studies in the seed corpus. This is a targeted-search yield, not a claim about the field's true publication trajectory.

Two further descriptive facts, both read directly from the coded database rather than estimated, complete the picture Figure 3 and Figure 4 give of *what kind* of hybridisation is reported: *where* in pavement engineering it concentrates, and *which* optimiser families do the coupling. Figure 5 shows both. Materials characterisation and performance prediction together account for just over half the corpus (38 studies each), consistent with §5 and §7's observation that these are the two sub-domains where a competing mechanistic or empirical model is most often already available for the tuned-baseline comparison §9 is built to test. On the optimiser side, a genetic algorithm or particle swarm optimisation account for over half of the 51 studies that name a specific optimiser family at all, with the remaining studies spread across more than twenty distinctly named metaheuristics appearing once or twice each — the empirical counterpart to the "optimiser proliferation" question posed in §13, and consistent with a field in which naming a new metaheuristic is, on its own, treated as sufficient novelty (§1).

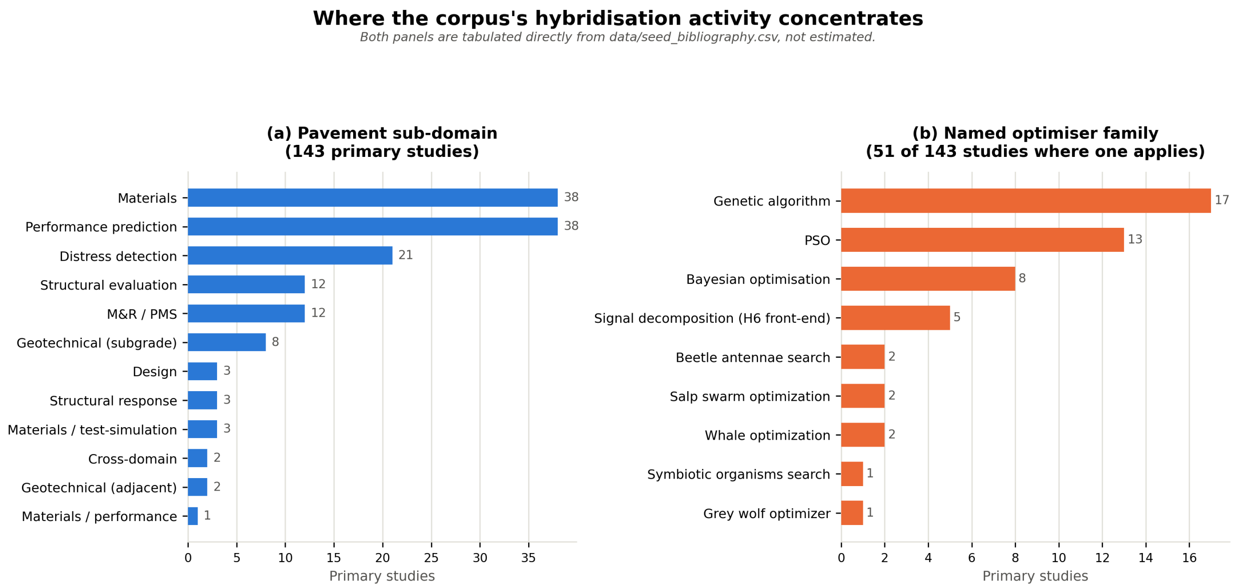

Figure 5: Where the corpus's hybridisation activity concentrates: (a) primary studies by pavement sub-domain, (b) primary studies by named optimiser family, among the 51 of 143 where a specific family applies. Both panels are tabulated directly from `data/seed_bibliography.csv`; panel (b) merges free-text spelling variants of the same optimiser (e.g. "PSO" and "particle swarm optimization") under one label, and excludes methods named only once each rather than folding them into an unrelated bucket.

4. Taxonomy of hybridisation

The field's literature treats "hybrid" as a single, undifferentiated label. This flattens together couplings with different theoretical justifications and, correspondingly, different characteristic failure modes — a metaheuristic searching a hyperparameter space is a different claim from a metaheuristic searching a weight space, and a different claim again from a physics-constrained loss function. Table 2 sets out the seven types this review distinguishes and codes against.

Table 2: Taxonomy of hybridisation in data-driven pavement models (H1–H7).

Type	Coupling	Theoretical justification	Characteristic failure mode
H1	Metaheuristic → hyperparameters	Searches a non-convex, discrete hyperparameter space that gradient methods cannot address	Compared against an untuned base learner, so the reported gain is the tuning, not the metaheuristic
H2	Metaheuristic → weights / structure	Escapes local minima of backpropagation; permits non-differentiable objectives	Vastly more function evaluations than the baseline receives, with no compute-parity reporting

Table 2: Taxonomy of hybridisation in data-driven pavement models (H1–H7).

Type	Coupling	Theoretical justification	Characteristic failure mode
H3	Physics + data (PINN, mechanistic residual, transfer function as loss)	Constrains the solution to a mechanically admissible manifold; aids under data scarcity	Physics-term weight tuned on the test set; constraint too weak to bind in practice
H4	Architecture fusion (e.g. CNN+transformer, two-stream)	Complementary inductive biases – local texture and long-range context together	Gains confounded with parameter count and training budget rather than the fusion itself
H5	Heterogeneous stacking / blending	Decorrelated errors across model families reduce variance	Meta-learner trained on in-fold predictions; leakage through the stacking layer
H6	Signal decomposition then learn (VMD, EMD, CEEMDAN, wavelet)	Separates scales the learner would otherwise have to disentangle unaided	Decomposition fitted on the full series before the temporal split – the most common severe leakage in this family
H7	Symbolic–numeric (GEP, MGGP, EPR + a numeric optimiser)	Yields a closed-form, inspectable expression rather than an opaque model	Expression complexity uncontrolled; an overfit equation is then read as if it were physics

Figure 6 gives the same seven types as a diagram, useful for distinguishing at a glance couplings that read similarly in prose — H1 and H2 differ only in whether the metaheuristic's output feeds a hyperparameter or a weight, H5 and H6 both fan multiple components into one learner but in the opposite direction (converging predictions versus converging decomposed inputs).

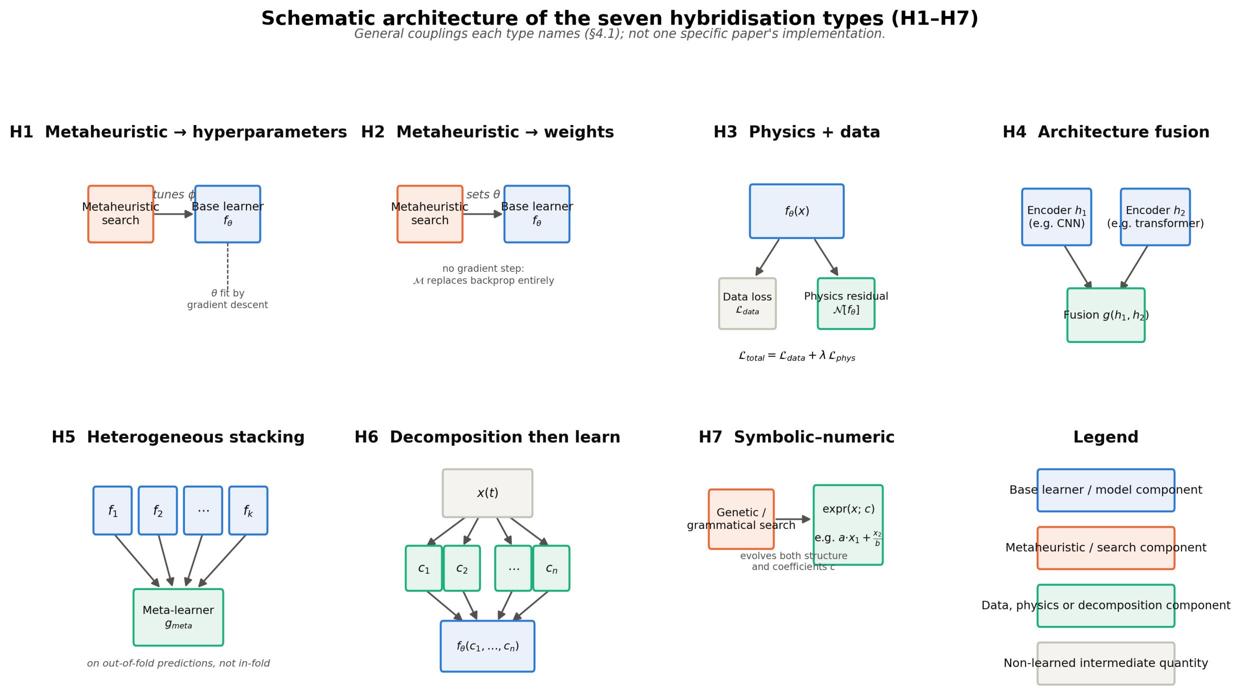

Figure 6: Schematic architecture of the seven hybridisation types. Boxes shown in the same colour play the same structural role across panels (base learner, search component, or data/physics/decomposition component); the diagram sets out the general coupling each type names, not any one paper's specific implementation.

4.1. Formal structure of each type

Naming each type by its coupling, as Table 2 does, leaves the coupling itself implicit. Stating it mathematically is what lets §9's premium construct be posed precisely for H1 and H2, and is done here for the remaining five types with the same notation throughout. Let f_θ denote a base learner with trainable parameters θ, x an input vector, and $\hat{y} = f_\theta(x)$ its prediction.

H1 — metaheuristic → hyperparameters. The base learner's hyperparameters $\phi \in \Phi$ (layer width, kernel bandwidth, tree depth, regularisation strength) are selected by a metaheuristic search rather than a grid or gradient method:

$$\phi^* = \arg\min_{\phi \in \Phi} \mathcal{L}_{\text{val}}\big(f_{\theta(\phi)}\big), \qquad \theta(\phi) = \arg\min_{\theta} \mathcal{L}_{\text{train}}(f_\theta \mid \phi)$$

with ϕ^* found by a population-based or trajectory-based search rather than by exhaustive or gradient-based enumeration of Φ.

H2 — metaheuristic → weights or structure. The metaheuristic instead searches the parameter space θ directly, replacing gradient descent:

$$\theta^{(t+1)} = \mathcal{M}\Big(\theta^{(t)},\ \mathcal{L}(f_{\theta^{(t)}}, X, y)\Big)$$

where $\mathcal{M}$ is the metaheuristic's own update operator — for particle swarm optimisation, for example, a velocity-and-position update over a population of candidate weight vectors — in place of $\theta^{(t+1)} = \theta^{(t)} - \eta\,\nabla_\theta \mathcal{L}$.

H3 — physics and data. A physics-informed loss term augments the data loss:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{data}}\big(f_\theta(x), y\big) + \lambda\,\mathcal{L}_{\text{phys}}\Big(\mathcal{N}[f_\theta](x)\Big)$$

where $\mathcal{N}[\cdot]$ is a mechanistic operator (a governing differential equation's residual, a transfer-function constraint) the trained function is penalised for violating, and λ weights the two terms against each other.

H4 — architecture fusion. Two encoders h_1, h_2 with different inductive biases are combined by a fusion function g:

$$\hat{y} = g\big(h_1(x), h_2(x)\big)$$

implemented as concatenation, cross-attention, or a learned gate, which is what distinguishes fusion from a single-family ensemble in which h_1 and h_2 share one architecture.

H5 — heterogeneous stacking. k base learners drawn from different families produce predictions a meta-learner combines:

$$\hat{y} = g_{\text{meta}}\big(f_1(x), f_2(x), \ldots, f_k(x)\big)$$

The leakage failure mode named in Table 2 is precisely when g_{meta} is fitted on in-fold predictions, values each f_i produced for inputs it had already seen during its own training, rather than on out-of-fold predictions generated under cross-validation.

H6 — decomposition then learn. A signal $x(t)$ is decomposed into n components before the learner is applied:

$$x(t) = \sum_{i=1}^{n} c_i(t), \qquad \hat{y} = f_\theta\big(c_1, c_2, \ldots, c_n\big)$$

where the c_i are intrinsic mode functions (EMD, CEEMDAN), wavelet coefficients, or VMD modes. The characteristic failure mode is the decomposition itself being computed over the full series, all t at once, before any train/test split is subsequently applied to it.

H7 — symbolic-numeric. An expression tree is evolved by a genetic or grammatical search (GEP, MGGP) over both its structure and its numeric coefficients c:

$$\hat{y} = \mathrm{expr}(x; c), \qquad \mathrm{expr}^* = \arg\max_{\mathrm{expr} \in \mathcal{T}} \mathrm{fitness}(\mathrm{expr}, X, y)$$

where $\mathcal{T}$ is the space of syntactically valid expression trees under the method's function and terminal sets — a structural search that distinguishes this type from H1 and H2, which search only over a fixed architecture's parameters or hyperparameters.

A representative metaheuristic, for concreteness. Particle swarm optimisation, the single most common search procedure named across the corpus, updates a population of candidate solutions x_i each iteration by

$$v_i^{(t+1)} = w\, v_i^{(t)} + c_1 r_1 \big(p_i - x_i^{(t)}\big) + c_2 r_2 \big(g - x_i^{(t)}\big), \qquad x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)}$$

where p_i is particle i's own best position found so far, g the swarm's best position overall, w an inertia weight, c_1, c_2 acceleration coefficients, and $r_1, r_2 \sim U(0, 1)$. Applied to H1, x_i encodes a hyperparameter vector ϕ; applied to H2, the same x_i encodes the base learner's own weight vector θ directly. The identical update rule, evaluated against a different loss surface, is what distinguishes the two types mechanically, while the structural test in §2.3 is what distinguishes them analytically.

This formal statement earns its place for one specific reason beyond notation: it makes concrete exactly what §9 asks a fair comparison to hold constant. For H1 and H2, the premium Δ (§9.1) compares $f_{\theta(\phi^*)}$ or f_{θ^*} against f_θ tuned by a conventional procedure under a comparable search budget — the same function f, reached by a different route to its parameters. A study that reports only the metaheuristic-tuned result has reported one point on this optimisation landscape without the second point that would say whether the metaheuristic, rather than the tuning itself, is what earned the reported gain.

Two coding notes follow directly from applying this taxonomy to the seed corpus. First, a single study can legitimately receive more than one code — [14] fits both a GA-derived symbolic equation (H7-adjacent) and an ANN whose weights are set by the same GA (H2) on the same data, which is precisely the kind of within-paper comparison §9 is built to exploit. Second, several studies that use metaheuristic or architecture-fusion vocabulary in their titles fail the structural test on inspection — [12] names both a genetic algorithm and an artificial neural network but reports them as independently competing models rather than a coupled pipeline, and is coded `none` rather than any of H1–H7 as a result. Both notes matter for the same reason: the taxonomy is applied to what a pipeline *does*, not to what its title or abstract *calls* it.

Reinforcement-learning approaches to network-level maintenance-and-rehabilitation policy [13, 10] sit adjacent to this taxonomy but are not coded into it: they couple a learned policy to a simulated or predicted environment in a way that does not map onto the single-prediction-pipeline structure H1–H7 assumes. They are retained as context for the decision-consequence discussion in §11 rather than folded into the hybridisation-type counts in §3.

5. Hybrid models in material and mix modelling

Material-property modelling is the sub-domain where the seed corpus shows the heaviest concentration of H1 (metaheuristic-tuned hyperparameters) and H7 (symbolic–numeric) activity, and it is also where the tuned-baseline problem identified in §9 is most visible, because the target properties — resilient modulus, dynamic modulus, unconfined compressive strength, air-void content — are exactly the properties for which competitor mechanistic or empirical models already exist and are routinely, though not universally, available for comparison.

The resilient-modulus literature illustrates both the promise and the recurring weakness. Ghorbani et al. [14] fit two related models on the same repeated-load triaxial dataset — a genetic-algorithm-derived symbolic equation and an ANN whose weights the same genetic algorithm sets (H2) — and report the hybrid ANN-GA model as the more accurate of the two, a rare instance in the corpus of a true head-to-head comparison between two hybridisation types on identical data rather than between a hybrid and an unspecified baseline. Azam et al. [2] instead compare six swarm-optimised variants of a single base learner (LSSVM) against one another; the top three separate by RMSE values of 6.72, 6.78 and 6.79 MPa and R^2 values of 0.942, 0.940 and 0.940 — a spread well inside what a different random seed would plausibly produce — and none is compared against a conventionally tuned, non-hybrid LSSVM, so no premium is computable from the paper as published (§9). Kaloop et al. [15] report a similar optimiser-held-constant, learner-varied design (PSO-ANN vs. PSO-ELM vs. kernel-ELM), useful for isolating base-learner choice but not, by itself, for isolating the optimiser's contribution.

The clearest positive exemplar in this section is not a hybrid at all in the H1–H7 sense: Zeiada et al. [1] individually tune eight learners, including deep architectures, and benchmark every one of them against the Witczak 1-40D and Hirsch mechanistic models for dynamic modulus. Under that design, the winner is a tuned bagging ensemble — not a hybrid and not a deep network — which is precisely the finding a hybridisation-premium framework should produce when tuning is held fair across all competitors, and it is the paper this review returns to repeatedly (§1, §9) as the standard the rest of the sub-domain should be read against.

Duan [16] supplies a genuinely mixed case worth reading in full rather than reducing to a single verdict, because full-text verification surfaces both a real strength and a real gap in the same paper. On the positive side, the metaheuristic-tuned BKA-XGBoost model (H1) reports its own search budget explicitly (population

10, 20 iterations) and, unusually for this corpus, checks its own reliability by reassembling the dataset ten times and reporting performance across all ten combinations rather than a single best run — exactly the repeated-run practice PAVE-ML item 12c asks for. The paper also states outright, discussing four comparator models drawn from the literature, that "direct performance comparisons are unconvincing" because those models were fitted on different datasets — an author-acknowledged version of the cross-study comparability problem this review is built around, and one of the only such acknowledgements found in the corpus so far. On the other side of the same paper: the nine within-paper comparator models are given listed parameter settings but no stated search budget, so whether they received tuning effort comparable to BKA-XGBoost's own cannot be confirmed from the text — and no mechanistic resistance-modulus model is included as a comparator, so baseline_strength codes moderate rather than strong despite the nine-model comparison looking exhaustive at a glance. The lesson is the same one §11 draws more generally: a paper can do real, citable work toward some PAVE-ML items while leaving others exactly as unaddressed as the rest of the field.

On the symbolic side, Ghanizadeh et al. [17] fit a TLBO-optimised evolutionary polynomial regression and multi-gene genetic programming model (H7) for asphalt air-void evolution, producing a closed-form, inspectable expression rather than an opaque predictor — a property that recurs as a theme in §10's discussion of what interpretability should mean for a validated, not merely a post-hoc-explained, model.

6. Hybrid models in structural evaluation and inverse analysis

Structural evaluation and inverse analysis — principally back-calculation of layer moduli from deflection data — is the sub-domain with the strongest example of external validation in the entire seed corpus, and for that reason anchors much of the argument in §11 as well as this section. Elbagalati et al. [18] train an ANN model on a Louisiana continuous-deflection programme and evaluate it on an independent Minnesota programme; R^2 falls only from 0.73 to 0.72 across that transfer, which is the benchmark against which the field's more typical 0.99-on-random-holdout results should be read, and the paper is cited repeatedly in this review precisely because it is the exception rather than the rule.

Ghanizadeh and colleagues' own back-calculation work traces a coherent five-year arc that this section follows in sequence: an ANN back-calculation model with Garson and connection-weight sensitivity analysis [19] establishes the baseline single-architecture approach with post-hoc interpretability; METABACKCAL [20] replaces the neural surrogate with a metaheuristic-driven inverse solver directly (H2, in the sense that the metaheuristic operates on the inverse problem itself rather than on a learned model's hyperparameters), and is released as a named tool — one of very few entries in the seed corpus with deployment evidence beyond a closed-form equation. Reading the two together raises a natural question this corpus is not yet positioned to answer at abstract level: does the metaheuristic-driven solver in the later paper report an accuracy gain over the earlier ANN surrogate under a comparable computational budget, and if so, of what size? Answering it would require full-text extraction from both papers rather than the abstract-level record currently held, and is flagged in §13 as a specific, tractable next step rather than a general aspiration.

Pasupunuri, Thom and Li [21] supply this review's clearest H3 (physics-informed) exemplar, embedding the MEPDG roughness model itself as the physics-loss term of a PINN for jointed plain concrete pavement roughness — the single clearest published case in the corpus of a mechanics-constrained learner in pavement engineering, and a useful counterpoint to the purely data-driven H1/H2 work that dominates the rest of this section.

7. Hybrid models in performance prediction

Performance and deterioration prediction (international roughness index, rutting, cracking, faulting, remaining service life) is the sub-domain most heavily dependent on the Long-Term Pavement Performance (LTPP) database in the seed corpus, and correspondingly the sub-domain where the near-duplicate-substrate problem named in the review methodology is most concrete rather than abstract. A single overlapping author team, Alnaqbi, Zeiada and Al-Khateeb, joined variously by Abuzwidah, Hamad and Barakat across different papers, has published at least nine studies located so far, all drawing on what appears to be the same approximately 33-section, 385–395-observation continuously-reinforced-concrete-pavement substrate from LTPP. Each pairs a different metaheuristic (chiefly PSO or a genetic algorithm) with a different base learner (chiefly SVR or a gradient boosting machine) to predict a different single target variable: international roughness index [22, 23, 24], longitudinal cracking [25, 26, 27], spalling [28], friction number [29], and punchouts [30]. Each is reported as a separate publication, each is evaluated by five-fold cross-validation, and each reports R^2 in the 0.90–0.99 range. Three pairs within this set differ from one another only in which metaheuristic tunes the same base learner, or which base learner the same metaheuristic tunes, on the same target and the same substrate: [25] (PSO-GBM), [26] (GA-GBM) and [27] (PSO-SVR) all predict longitudinal cracking; [22] (PSO-SVR), [23] (PSO-GBM) and [24] (GA-SVR) all predict IRI on what the papers describe as the identical 395-observation, 33-section pool. The most recent addition, [27], is inferred to belong to the same series on topical and authorship grounds (the same three authors, the same rigid-pavement cracking target, the same optimiser-learner coupling pattern) rather than confirmed against its own data section, which full-text reading would be needed to settle with certainty.

None of this is concealed. [30]'s own reference list cites six other papers from the same series, so the pattern is not hidden, only undiscussed. Full-text verification of the first two published members of the series, [22] and [25], confirms rather than merely suspects the leakage concern, and confirms it identically in both papers rather than once. The first states its partitioning plainly as "the dataset was split into 5 subsets, and the model was trained and tested 5 times, each time using a different subset as the test set and the remaining subsets as the training set"; the second, on a different target variable but the identical

395-observation, 33-section substrate, states it just as plainly as “five equal-sized subsets were randomly selected from the dataset… the training and testing subsets were switched around in each iteration.” Both are random splits at the observation level, and neither methods section, read in full rather than excerpted, mentions grouping by the 33 physical CRCP sections the observations are drawn from. With roughly twelve observations per section on average, an ungrouped random split places multiple, highly correlated records from the same physical section on both sides of most folds: precisely the leakage pathway the PAVE-ML rule for `leakage_risk` (§12) is written to catch. Both papers’ own Limitations or Future Directions sections separately confirm the absence of external validation in their own words (the first stating outright that “no independent testing was done using additional datasets,” the second listing “external dataset validation” among unmet future work), which independently confirms the `external_validation: no` coding under the criterion in §2.3 twice over, rather than by the silence-as-absence convention alone. One detail is worth noting for what it reveals about how such concerns are currently addressed in this literature: the first paper anticipates an overfitting objection and defends against it by observing that RMSE stays within a narrow band across folds (0.02791–0.06422). But fold-to-fold consistency is not evidence against the leakage pathway just described, since every fold would share the same section-level contamination and so would be expected to perform similarly whether or not the model has learned anything section-specific. Figure 7 makes the mechanism concrete: with roughly twelve observations per section, an ungrouped random assignment (Panel A) all but guarantees that most sections contribute records to more than one fold, so a model can be trained on 10–11 of a section’s 12 observations and then “tested” on the 12th, a near-duplicate, spatially correlated record rather than a genuinely unseen one. A section-grouped assignment (Panel B), which the reviewed papers’ methods sections do not describe using, holds each section entirely within one fold, so the test fold contains no information the model could have seen during training.

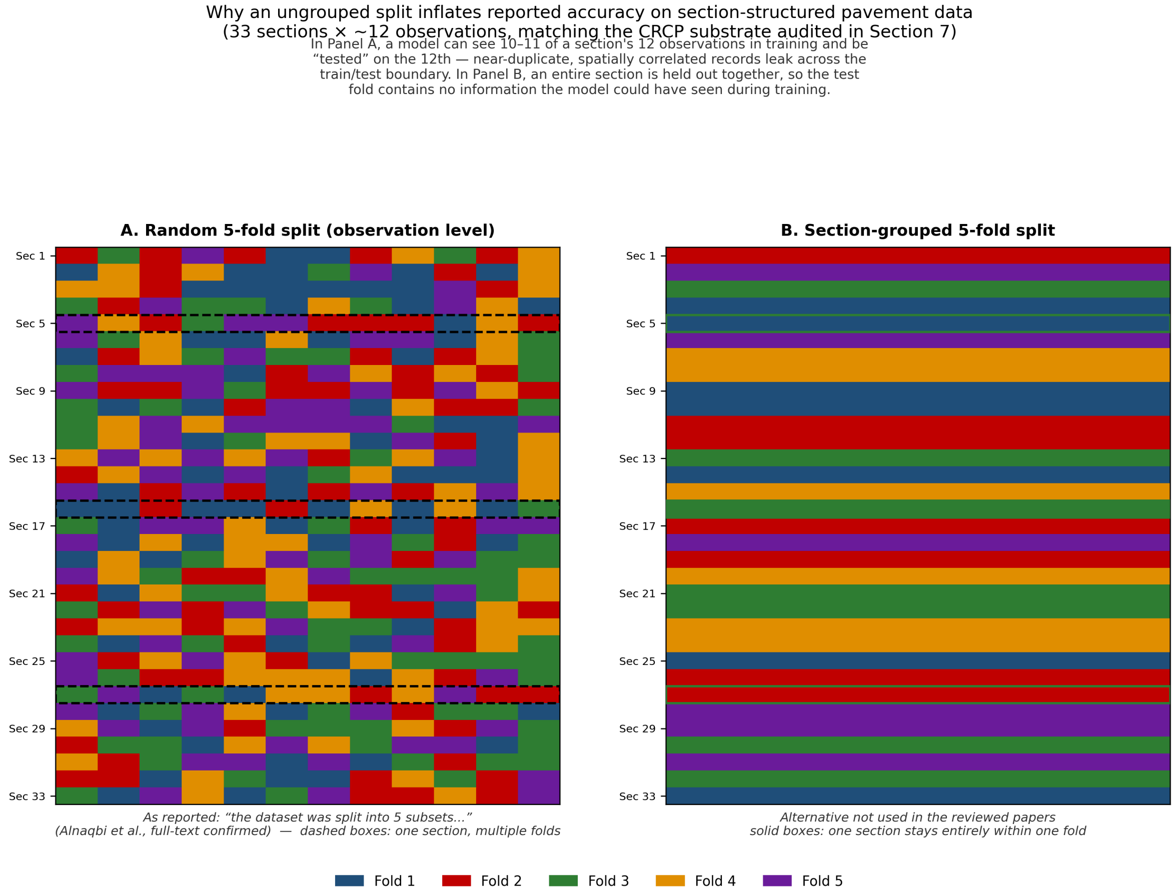

Figure 7: Random versus section-grouped 5-fold cross-validation on a 33-section, ~12-observation-per-section substrate matching the CRCP data audited in this section. Panel A reproduces the partitioning [22]'s methods section describes; Panel B shows the section-grouped alternative not used in the reviewed papers.

Both papers also describe z-score normalisation, imputation and outlier removal within a single "data preprocessing" stage that textually precedes the cross-validation description, with no statement in either that these steps were refit inside each fold rather than computed once over the full pool. The remaining six papers in the series are coded `high` risk pending the same full-text check now completed, identically, for the first two; nothing in their abstracts suggests a different partitioning procedure was used, and two independent confirmations make that a considerably stronger prior than one.

Two further pieces of evidence extend the pattern beyond one author team, and one of them is now full-text confirmed rather than ambiguous. Xiao et al.'s TPE-CatBoost faulting model [31], read in full, resolves its own ambiguity definitively. The paper's Boruta feature-selection step is performed in Section 3 ("Data Preparation"), applied to the full 160-observation pool, and only in Section 4 ("Model Construction") does an 80/20 train-test split appear, meaning Boruta selected the six variables the final model uses with the benefit of information from what later became the test partition. This is a different leakage mechanism from the CRCP series above: a real held-out test set exists here, contaminated only by feature selection performed before the split, rather than an entirely ungrouped cross-validation. The paper's own conclusion attributes part of its headline result directly to the contaminated step, reporting that the six-variable model improves R^2 by 0.007 over the seventeen-variable version, "attributed to the capability of Boruta to identify relevant variables." The two mechanisms, read together, make the point this review is built to make more sharply than either alone: `leakage_risk` in PAVE-ML (§12) catches structurally different failures (no partition grouping in one case, preprocessing fitted before the partition in the other) under a single item, which is exactly why the item is framed around the outcome (does information from the test set influence training) rather than around one specific mechanism. A citation search while verifying this paper separately surfaced a second, seemingly independent overlapping-author cluster: Wang, Xiao and Liu [32], sharing two of the same three surnames as the TPE-CatBoost paper, publishing a separate improved-XGBoost model for rigid pavement IRI on what appears to be a related LTPP substrate. Whether this reflects a second, independent instance of the single-target-per-paper pattern, or overlapping personnel with the first cluster, is not resolved by this review's corpus and is flagged in §13 as a specific question a wider, full-text-verified sweep could settle.

Two things distinguish this section's treatment of the near-duplicate-substrate problem from a general methodological aside. First, the count is large enough — nine identified papers, two of them full-text-confirmed rather than only abstract-flagged — that this review can report a systematic, partly-verified finding rather than gesture at a concern: of the two papers read in full, both use an ungrouped observation-level split on the shared substrate, which is already sufficient to establish that the pattern is real and not merely a plausible-sounding hypothesis, even though the remaining seven papers in the series have not yet been read with the same care. Second, the pattern usefully illustrates a broader point this review returns to in §11: an unlimited supply of near-identical papers reporting R^2 above 0.90 on the same underlying 33 physical objects, whatever its cause, produces exactly the appearance of a mature and thoroughly validated sub-field that the rigour audit is designed to test.

Deng and Shi's practitioner-facing review of rutting models [33] adds a data point this review's own audit cannot generate on its own: a survey of which rutting-prediction models highway agencies actually report using in practice. That detail belongs with the deployment-evidence discussion in §9 and §12 as one of the only direct looks, anywhere in the surrounding literature, at the demand side of the deployment gap this review otherwise infers only from what published papers do not report.

8. Physics-informed, fusion, stacking, decomposition and symbolic hybrids

The five hybridisation types beyond direct metaheuristic-hyperparameter coupling (H3–H7) are less numerous in the seed corpus than H1, but each supplies at least one clear exemplar worth setting out in detail because each illustrates a distinct failure mode that a reader unfamiliar with the taxonomy in §4 would not otherwise anticipate. This asymmetry is worth naming plainly rather than only implying: H1 (37 primary studies, §3/Figure 3) is substantially better sampled than H3, H4, H5 or H6 individually (4–11 records each), partly because the field genuinely produces more H1 studies and partly because H1 and H7 titles are unusually easy to find and code with confidence from a title alone (§2.3), which biases what an algorithm-driven search protocol surfaces first. The characteristic-failure-mode claims made below for H3–H7 are drawn from real, verified records, not fabricated, but a claim resting on a handful of records carries less evidentiary weight than one resting on several dozen, and is presented accordingly: as a documented pattern within a substantial but not exhaustive sample, not as a settled distributional fact about the field as a whole.

H4, architecture fusion, is best represented in the seed corpus by four vision and one network-level study discussed in detail below, plus four further CNN–transformer crack-detection studies added from a 2026 harvest pass [34, 35, 36, 37] that were search- or title-confirmed rather than full-text verified — together they show the CNN+transformer fusion pattern recurring from 2023 through 2026 rather than appearing once. The best-cited of the four, Guo et al.'s Swin-Transformer-encoder, attention-augmented-UperNet-decoder network [37], reports 102 citations at the time of writing — more than any other H4 record in the corpus — and an explicit within-paper comparison against CNN-only baselines, which is the staged-ablation pattern this section otherwise has to go looking for. Sun et al.'s PCDETR [38] fuses a Swin-Transformer and a ResNet backbone for pavement crack detection and reports near-identical average precision on the authors' own dataset and on COCO (45.6% vs. 45.8%) — a similarity close enough across two very different image domains to warrant scrutiny rather than to be read at face value as evidence of robust transfer. Alshawabkeh et al. [39] and Elhariri et al. [40] both couple deep and hand-crafted feature extraction with a metaheuristic feature-selection stage (jointly H1 and H4 under the taxonomy in §4), and both report a staged, within-paper ablation — deep features alone, then with the added classifier, then with the added optimiser — which is the design §9's premium construct depends on and which most H1 studies elsewhere in the corpus do not provide. At the network rather than the image level, Pan, Du and Parlikad's PaveGNet [41] fuses a graph module encoding inter-section topology with a temporal module encoding distress evolution, and is, at the time of writing, this review's clearest example of the staged-ablation design done well: independently substituting the temporal, spatial and exogenous-variable components and reporting the ranking-error contribution of each (a fall from 9.005% to 2.670% for the full model against a spatiotemporal graph-convolution baseline). It is a useful positive counterpoint to place beside the H1 studies earlier in this section, and is the closest the seed corpus comes to network-level, rather than single-section, hybridisation evidence.

H5, heterogeneous stacking, has one clear positive and one ambiguous exemplar in the corpus, both already introduced in §9: Huang et al. [42] train their meta-learner on cross-validated rather than in-fold base-model predictions specifically to avoid leakage — full-text verification confirms the design does what its abstract claims, with a formal description of the fold structure matching a textbook leakage-safe stacking implementation — and is cited throughout this review (§9, §12) as the type's positive case; Yu et al. [43] stack five heterogeneous learners for tunnel pavement performance within a digital-twin framework and report strong deployment integration, but without the explicit leakage-avoidance statement that distinguishes Huang et al.'s design. Four further stacking studies from the 2026 harvest pass [44, 45, 46, 47] are coded H5; the last of these is search-confirmed rather than title-only — Peng et al. name their base and meta-models explicitly (ridge regression, k-NN, MLP and random forest stacked under an SVM meta-learner) and report lower variance and bias for the stacked model than for the individual base models, though the exact comparator protocol still needs full-text confirmation before that becomes a quoted premium figure. None of the four has yet been checked for which side of the Huang/Yu leakage-practice split it falls on — this is exactly the kind of question the full-text verification queue exists to answer, not something to guess from a title.

H6, decomposition then learn, was reported through the previous phase of this review as the one taxonomy type with no confirmed pavement-specific exemplar in the seed corpus, after two independent, differently worded search sweeps failed to find one. That was an honest report of what those two sweeps actually covered, not a structural claim about the literature, and it has not survived a third, larger pass: a 2026 harvest search identified Kaloop et al.'s wavelet–OP-ELM model for rigid pavement roughness [48], whose own abstract states plainly that "Optimally Pruned Extreme Learning Machine (OP-ELM) and Wavelet analysis are integrated" and benchmarks the coupling against plain OP-ELM, a conventional ANN and a regression baseline on LTPP data — a genuine, abstract-confirmed H6 case. A second, more tentative candidate [49] pairs ensemble empirical mode decomposition with a downstream predictor for pavement modulus, though the retrievable abstract snippet does not name that downstream model explicitly, so it is coded provisionally pending full-text confirmation. Neither record's leakage risk is known: both publishers blocked automated full-text access — the same failure mode recorded against a separate, earlier verification attempt on a ScienceDirect-hosted paper (documented in the database, not repeated here) — so whether the decomposition step was fitted before or after the train/test split — the exact H6 failure mode this taxonomy names — remains unread rather than assumed either way. A follow-up search pass the same day found two further, older H6 records that a single query had missed: Nejad and Zakeri's paired 2011 studies [50, 51] each couple a wavelet-domain decomposition (wavelet–Radon transform in one, wavelet transform alone in the other) to a distinct trained neural-network classifier for pavement distress recognition — the same two authors, the same journal and year, two related but architecturally different H6 designs rather than one. H6 in this corpus is therefore no longer a single exemplar but a small evidence base of four records spanning 2011 to 2025, which changes what can honestly be said about it: not "does H6 exist in pavement engineering" (yes), but "how often is it done safely," which full-text verification of these four is now positioned to answer. A fifth, superficially similar record is worth naming precisely because it is *not* H6: Ayenu-Prah and Attoh-Okine's bidimensional empirical-mode-decomposition study [52] — at 170 citations, the best-cited record this pass touched — pairs BEMD with a Sobel edge detector, a fixed classical operator, not a trained model, so it fails the structural test in §2.3 despite matching every keyword an automated screen would use to flag it as H6. It is the cleanest available illustration of the type's boundary: decomposition followed by a fixed operator is not H6; decomposition followed by something that learns from data is. The corrected finding is therefore that H6 is rare but not absent, that its absence in earlier phases reflected search coverage rather than a property of the literature, and that the field's own vocabulary ("wavelet-based," "EMD-based") does not reliably distinguish the two cases — exactly the vocabulary problem §1 opens with, recurring one taxonomy type later. The research agenda in §13 is updated accordingly.

H3 (physics-informed) and **H7 (symbolic–numeric)** are each introduced in §6 and §5 respectively [21, 17, 14] and are not repeated here in full; both recur in §10's discussion of what a validated, mechanically consistent model should be able to claim about its own interpretability that a purely post-hoc explanation of an unvalidated model cannot.

9. The hybridisation premium

9.1. Definition

For a study reporting a hybrid model M_{hyb} (optimiser $\oplus$ base learner M_{base}) evaluated on a held-out partition, the **hybridisation premium** is

$$\Delta = \text{performance}(M_{\text{hyb}}) - \text{performance}(M_{\text{base}}^{\text{tuned}})$$

where $M_{\text{base}}^{\text{tuned}}$ is the same base learner, tuned by a conventional procedure under a **comparable evaluation budget**, evaluated on the **same** partition with the **same** metric. All three constraints are load-bearing: a premium computed against an untuned, default-setting baseline is not a measure of what hybridisation contributes — it is a measure of what tuning contributes, mislabelled.

Operationalising "comparable evaluation budget." A grid search, a random search and a population-based metaheuristic (PSO, GA, and the many named variants in §4) do not share an obvious common unit — a grid search has no generations, a metaheuristic has no grid resolution. The unit this review uses is the one quantity every search strategy actually shares: **the number of times the base learner itself is trained and evaluated during the search** — total grid points for a grid search, total random draws for a random search, population size × number of generations (or the paper's own reported total function-evaluation count) for a metaheuristic. Two arms are treated as budget-comparable when their base-learner training-and-evaluation counts are within the same order of magnitude; wall-clock time is used as a secondary check where reported, since it can diverge from the evaluation count when the optimiser itself carries non-trivial overhead. This is also what PAVE-ML item 12b (§12) asks studies to report directly, so that this determination does not have to be inferred indirectly from a paper's stated hyperparameter ranges. A study that tunes the hybrid arm exhaustively while leaving the baseline arm at library-default settings fails this test regardless of which search algorithm names appear in its title — the H1/H2 failure mode both the taxonomy in §4 and the evidence in §9.3 keep surfacing.

9.2. Coding scheme for premium evidence

Every H1 or H2 study in the corpus is coded against three fields, specified in full with worked examples in the PAVE-ML instrument (§12) and fixed before coding began so that later readings could not drift toward a preferred answer: whether Δ is computable at all under the definition above (`premium_computable`); its value where it is; whether the search budget was stated for both the hybrid and the baseline arm (`budget_parity_reported`); and, where a newly named optimiser is the paper's own contribution, whether it was benchmarked against at least two established optimisers on the same problem (`optimiser_novelty_claim`).

9.3. Evidence from the coded corpus

Table 3 collects the within-paper comparisons in the corpus close enough to a premium calculation to be directly informative, with every figure copied verbatim from the source paper's own reported results; none is computed or estimated by this review.

Table 3: Within-corpus evidence bearing on the hybridisation premium. Every figure is copied verbatim from the cited paper's own reported results; none is computed by the present review.

Source	Comparison	Metric	Non-hybrid value	Hybrid value	Premium	Note
`10.3390/app9153172`	Plain SVM vs. GA-ANFIS vs. PSO-ANFIS, all reported within one paper, 1000-run Monte Carlo	RMSE / MAE / R (Marshall Stability)	plain SVM (best of three by converged statistical criteria)	GA-ANFIS and PSO-ANFIS (both hybrids)	negative — plain SVM outperforms both hybrids	Rare in this literature: a statistically robust (1000-simulation Monte Carlo) same-paper comparison where a plain, non-hybrid learner beats two separate metaheuristic-hybrid variants. Directly supports the review's central claim rather than merely gesturing at a missing baseline.
`10.1038/s41598-022-17429-z`	LSSVM (best single-kernel setting reported) vs. best of 6 swarm-optimised LSSVM variants	R2 / RMSE (MPa)	not reported as a standalone tuned baseline	0.942 / 6.72 (best of six, SOS)	not computable — no tuned non-hybrid baseline reported	The paper compares six hybrids to each other, never to a conventionally tuned LSSVM. This is the missing-baseline pattern Section 9 quantifies.

Table 3: Within-corpus evidence bearing on the hybridisation premium. Every figure is copied verbatim from the cited paper's own reported results; none is computed by the present review.

Source	Comparison	Metric	Non-hybrid value	Hybrid value	Premium	Note
`10.28991/cej-2025-011-01-06`	Eight individually-tuned learners (ANN, RNN, CNN among them) vs. Witczak/Hirsch mechanistic baselines	R2 (dynamic modulus)	Witczak 1-40D / Hirsch (mechanistic)	bagging ensemble, individually tuned (highest of eight)	positive but modest — mechanistic models remain competitive; deep architectures do NOT win	The field's clearest example of tuning parity: every learner gets its own search budget, and the winner is a tuned ensemble tree, not a hybrid or a deep net.
`10.1016/j.sandf.2020.02.010`	GA-fitted symbolic equation (H7-adjacent) vs. ANN-GA (GA on ANN weights, H2)	R2 (resilient modulus)	GA-only symbolic model	ANN-GA hybrid	ANN-GA reported higher R2, but adds a black-box layer over an already-competitive symbolic model	Rare true head-to-head between two hybrid types on identical data — flagged for full-text extraction of exact R2 values.

Table 3: Within-corpus evidence bearing on the hybridisation premium. Every figure is copied verbatim from the cited paper's own reported results; none is computed by the present review.

Source	Comparison	Metric	Non-hybrid value	Hybrid value	Premium	Note
`10.3390/app9163221`	PSO-ANN vs. PSO-ELM vs. kernel-ELM, all reported within one paper	RMSE / R2	PSO-ANN (same optimiser, different base learner)	PSO-ELM (best reported)	small, same-optimiser architecture comparison — isolates base-learner choice, not optimiser value	Useful for a different question than the premium: it holds the optimiser constant and varies the learner, the mirror image of what Section 9 needs.
`10.3390/ma18122913`	Single ML models (KNN, Bayesian ridge, decision tree) vs. stacking ensemble	R2 (G* and phase angle)	best single model	stacking (R2 = 0.973 / 0.999)	positive, and the ONLY row in this table where leakage is explicitly controlled by design	Cross-validated meta-features stated as a deliberate leakage-avoidance choice — the paper we cite as the positive PAVE-ML exemplar.

Three readings of this table matter more than any single row. First, **the comparison the premium requires is usually absent, not unfavourable.** Of the six rows, only three [2, 42, 53] involve a paper that reports enough to attempt a premium calculation at all, and [2]'s own six-optimiser comparison never includes the one comparator — a conventionally tuned, non-hybrid version of the same LSSVM — that would let a reader compute Δ. This is the pattern the full audit is built to quantify: the missing baseline, not a negative result, is the dominant finding so far. Second, **where tuning parity is closest to being honoured, the hybrid or deep-architecture advantage narrows — and can reverse outright.** [1] tunes eight architectures individually, including deep networks, against both each other and two established mechanistic models; the winner is a tuned bagging ensemble. [53] goes further: across a 1000-run Monte Carlo comparison, a plain support vector machine outperforms both a GA-tuned and a PSO-tuned ANFIS hybrid on the identical data — one of the seed corpus's few statistically robust cases where hybridisation is outright negative rather than merely unproven. Neither a hybrid nor a deep architecture prevailed in either study once every competitor received its own search budget. Third, [42] is, so far, the seed corpus's clearest positive counter-example — a heterogeneous stacking ensemble (H5) that both reports a credible premium over single models and states explicitly that its meta-learner was trained on cross-validated, not in-fold, base-model predictions specifically to avoid the leakage pathway characteristic of that hybridisation type (§4, H5). It is cited in §12 as the positive PAVE-ML exemplar for exactly this reason.

None of these readings should be generalised past what six studies can support. They are reported here as the reason the systematic audit in this section is worth completing, not as its substitute. It is worth noting, too, that this review is not alone in noticing the shape of the problem even if no prior source has quantified it the way this section does: a narrower, CBR-specific AI review published concurrently with this one independently argues that combining a base learner's own hyperparameters with an optimisation algorithm's hyperparameters adds a configuration complexity that can itself degrade accuracy [54] — a different diagnosis (configuration complexity rather than an unfair baseline) converging on the same underlying concern from a different sub-literature.

10. Interpretability and uncertainty in hybrid pipelines

Two constructs recur across the hybrid-model literature under the banner of trustworthiness — interpretability and uncertainty quantification — and both are applied in this corpus in ways that, on inspection, often do less than their presence in a paper's keyword list implies.

10.1. What "interpretability" means for a hybrid pipeline

The interpretability methods visible in the seed corpus fall into two families with different epistemic status. The first is **post-hoc attribution on an opaque model**: Garson's algorithm and connection-weight decomposition [19], SHAP applied to gradient-boosted or stacked ensembles [31, 42], and neighbourhood component analysis [55]. These methods answer a well-defined but narrow question — which inputs moved this particular model's output, on this particular data — and answer nothing at all about whether the model's predictions are trustworthy off that data. The second is **intrinsic interpretability**: symbolic-regression methods (GEP, MEP, EPR, MGGP) that yield a closed-form equation rather than a black-box function [56, 17, 57]. An equation can be inspected for physical plausibility — does the sign of each coefficient match known mechanics, does the functional form saturate or diverge where it should — in a way a SHAP plot cannot be, because SHAP explains what the model learned, not whether what it learned is correct.

This distinction sharpens the point §7 and §9 make about validation. An attribution method applied to a model that has not been externally validated (the majority of the corpus, by the audit's own count) ranks the structure of that model's training data, not the physics of the pavement — and every one of the corpus's SHAP applications identified so far [31, 42] is computed on a model whose external validation, under the criterion in §2.3, codes `no`. This is not a claim that the attributions are wrong; it is a claim that their scope is narrower than the surrounding prose typically implies. A useful negative case makes the point concretely: [58] apply SHAP to a two-predictor roughness model (age and cumulative traffic) — with only two inputs, a SHAP decomposition can say little a simple partial-dependence plot would not already show, and the paper illustrates how readily explainability tooling is reached for even where the underlying model offers little for it to explain. At the other end, [57] make the trade-off explicit rather than leaving it implicit: their ANN outperforms GEP and MEP numerically, and they recommend the symbolic alternatives anyway, stating plainly that this is "due to the blackbox nature of the ANN" — one of the only studies in the corpus to treat interpretability as a design objective competing with accuracy, rather than as a figure added after the fact.

10.2. Uncertainty quantification: absent, not merely under-reported

Where interpretability is present but often narrow, predictive uncertainty is, so far, close to absent. None of the studies coded to date reports an interval or distributional estimate for an individual prediction — no conformal intervals, no deep ensembles, no Gaussian-process variance, no quantile regression. What the corpus does report, frequently, is dispersion of the *performance estimate* across cross-validation folds (§9's discussion of [22]'s fold-consistent RMSE is one example), which is a different quantity: it describes how stable the model's average skill is across resampling, not how confident any single prediction should be treated. The two are easy to conflate and the corpus's papers do not, so far, distinguish them explicitly. Reporting this distinction is itself one of the audit's findings — a field that quantifies dispersion of its own performance estimate but not of its individual predictions has a form of uncertainty reporting that looks more complete than it is.

10.3. Physics-informed learning as a third, structural, path

[21]'s PINN offers a genuinely different route to trustworthiness than either post-hoc attribution or a closed-form equation: rather than explaining a model after fitting or restricting its functional form, constrain what the model is permitted to learn to what the underlying MEPDG mechanics allow, expressed as a physics-loss term during training. Sensitivity analysis is reported as a supplementary check, not the primary

trust mechanism — the mechanics constraint does that work directly. This is the only example of this kind in the seed corpus (§8), and its scarcity is itself the finding: a pavement engineering literature that has adopted post-hoc explanation tools readily has adopted structural, mechanics-consistent alternatives far more slowly, despite the domain's underlying physics being comparatively well characterised relative to many fields where PINNs originated.

11. From premium to decision

A premium, once measured, still needs to be worth having. Two studies, from either end of the pavement management hierarchy, supply a yardstick for judging that. Hosseini and Smadi [59] work at the network level: injecting 10–90% error into a pavement performance prediction and propagating it through a 20-year network-level maintenance and rehabilitation schedule, they price the resulting treatment-cost consequence directly. Georgouli, Plati and Loizos [60], newer and at the project level, price the same kind of consequence for a single input rather than a whole prediction: they show that the choice between the NCHRP 1-37A dynamic-modulus model and a locally calibrated alternative propagates through a mechanistic-empirical rutting simulation (3D-Move) into a materially different rutting estimate, with the uncalibrated model systematically overestimating deformation. Read together, the two studies supply a yardstick for a hybridisation premium at both ends of the pavement management hierarchy — not whether $\Delta > 0$, but whether Δ is large enough, at the accuracy level the underlying prediction task already operates at, to change which treatment gets scheduled, when, or how a single input propagates through a downstream design calculation. A premium of a few hundredths of a unit of R^2, of the kind several rows in Table 3 report or gesture toward, is a strong candidate for being decision-irrelevant by this standard. Making that judgement study by study, rather than asserting it in general, requires the kind of domain-specific decision context (which treatment, at what accuracy threshold, informed by which input) that most studies in this corpus do not state explicitly enough to apply the Hosseini–Smadi or Georgouli–Plati–Loizos yardstick directly; §13 identifies this systematically, rather than case by case, as a research-agenda item in its own right.

A related and more immediate question is how often the gap between training and test performance is even visible enough to judge, and here one paper in the seed corpus supplies a rare, self-reported answer rather than requiring an external audit to surface it. Jalal et al. [61] report four metaheuristic-tuned ANN variants (PSO, GWO, SMA and MPA) for expansive-soil swell pressure and unconfined compressive strength, with training R^2 above 0.9 for the UCS models — and then state plainly, in their own results section, that "the UCS models witnessed an overfitting problem because the aforementioned R values of the metaheuristics were 0.62, 0.56, and 0.58" on the test partition. This is, so far, the only paper in the corpus to name its own train-test gap in these terms rather than reporting a single headline R^2 and moving on, and it is a useful reminder that the rigour audit's purpose is not to catch something no author could have noticed — the gap is visible in the paper's own tables — but to make visible-but-unremarked-upon gaps like this one a counted, systematic finding rather than an exception a careful reader happens to spot.

12. PAVE-ML: a reporting and appraisal checklist

The audit this review conducts is only as trustworthy as the coding rules behind it, and those rules are only useful to the field if they survive as something other than this paper's internal methodology. PAVE-ML is offered as both: the instrument this review scores every included study against, and a checklist future studies can score themselves against before submission. Table 4 reproduces the full instrument; the operational decision rules and worked examples that make each item's coding reproducible rather than a matter of reader judgement are given in full in the supplementary material (`docs/03_PAVE-ML_instrument.md`).

Table 4: The PAVE-ML checklist: 20 general items across five domains, plus 4 items specific to hybrid pipelines.

#	Item
D1. Data provenance and structure	
1	Origin of every record stated (lab programme, named field project, named public database, or simulation), with acquisition period.
2	Unit of observation defined; paper states whether records are independent (e.g. repeated measurements on one specimen/section/image).
3	Sample size reported per split; ratio of records to fitted parameters or candidate inputs is computable from the text.
4	Input and target ranges given; paper states predictions outside those ranges are not supported.
D2. Preprocessing and partitioning	
5	Split described precisely enough to reproduce: mechanism, proportions, and random seed (or its absence).
6	Every fitted preprocessing step (scaling, imputation, resampling, feature selection, augmentation) stated to be fitted inside the training partition only.

Table 4: The PAVE-ML checklist: 20 general items across five domains, plus 4 items specific to hybrid pipelines.

#	Item
7	If records are non-independent (item 2), the split respects the grouping – no specimen/section/image/site appears in more than one partition.
8	Class imbalance, censoring, or target skew reported, with its treatment described.
D3. Modelling and comparison	
9	Architecture fully specified: layers, widths, activations, regularisation; base learner and count for ensembles.
10	Hyperparameter search described (space, method, budget); the selection partition is named and differs from the reported test partition.
11	At least one baseline is a tuned conventional model; the established domain/mechanistic model is included as a comparator where one exists.
12	Where a metaheuristic is used, whether it optimises hyperparameters or weights is stated, with a comparison against a conventionally tuned version of the same learner.
12a *(hybrid-specific)*	Base learner reported both with and without the hybridising component, on the same partition and metric.
12b *(hybrid-specific)*	Search budget stated for both hybrid and baseline arms (function evaluations, iterations × population, or wall-clock time).
12c *(hybrid-specific)*	Stochastic optimisers run more than once; spread across runs reported, not only the best run.
12d *(hybrid-specific)*	A newly named optimiser is compared against at least 2 established optimisers on the same problem; the No-Free-Lunch implication is acknowledged.

Table 4: The PAVE-ML checklist: 20 general items across five domains, plus 4 items specific to hybrid pipelines.

#	Item
D4. Evaluation, uncertainty and interpretation	
13	Metrics suit the task (e.g. IoU/F1 with the positive class defined for imbalanced segmentation, never bare pixel accuracy; an absolute-error metric in engineering units alongside R^2 for regression).
14	Performance reported for training AND held-out partitions, with dispersion (fold SD, CI, or repeated-run range), not a single point estimate.
15	Predictive uncertainty quantified for individual predictions (conformal intervals, deep ensembles, MC dropout, GP variance, quantile regression) or its absence acknowledged.
16	Any interpretation output (SHAP, PDP, Garson, connection weights, NCA, attention) accompanied by the explained model's validation status and a mechanics-consistency check.
D5. Generalisation, transparency and use	
17	Evaluated on at least 1 dataset from an independent source (different agency/region/lab/test track/campaign) or the absence of external validation is stated as a limitation in the conclusions.
18	Code and trained weights available at a persistent identifier; data available or the barrier to release named.
19	Computational cost reported: training time, inference time per record/image, and hardware.
20	Intended decision context stated: which engineering decision the output informs, at what level (project/network), and what accuracy that decision requires.

Two design choices are worth stating here because they determine how the audit's headline numbers should be read. First, **silence is coded as absence, not as unclear** — if a study does not state that its test partition was held out independently of hyperparameter selection, that is coded `no`, following the convention used in clinical prediction-model appraisal (TRIPOD, PROBAST), because it is the only convention that cannot be accused of charitable reading. Second, the four hybrid-specific items require a study to report its base learner's performance both with and without the hybridising component on the same partition and metric (item 12a); to state the search budget for both arms so that parity can be assessed (12b); to run stochastic optimisers more than once and report the spread rather than the best run (12c); and, where a newly named optimiser is the contribution, to benchmark it against at least two established optimisers on the same problem (12d). [42] is, and is now full-text confirmed to be, the clearest example in the seed corpus of a study satisfying the spirit of items in this list without having been written against it — its stacking meta-learner is trained on cross-validated, not in-fold, predictions specifically to avoid leakage, stated explicitly in the paper's own methods section rather than inferred, which is exactly the practice item 7 of the general instrument (grouped/blocked partitioning respected by all fitted preprocessing) is written to detect the absence of. One caveat keeps this a partial rather than a complete positive case: the same paper's external validation still codes `no` under the strict criterion in §2.3, since all data come from the authors' own laboratory tests rather than an independently sourced dataset — a reminder that a study can get one PAVE-ML dimension right while another remains open, and that the instrument is built to report per-item findings rather than a single pass/fail verdict per paper. The mirror-image worked example — full-text confirmation of the failure mode item 7 is written to catch, rather than of its avoidance — is the pair [22] and [25] (§7): both papers' methods sections state their five-fold splits plainly as random partitions of individual observations, with no grouping by the 33 physical sections those observations are drawn from, and both papers' own Limitations or Future Directions sections separately confirm the absence of external validation, independently of each other. The three papers, read together, are what the fully coded corpus is expected to look like at scale: a minority explicitly designing around the leakage and validation items in this instrument, and a majority whose methods sections, once actually read rather than inferred from an abstract, turn out to say plainly what §2's silence-as-absence convention would in any case have coded — and, in the case of the two full-text-confirmed papers, say it identically to one another despite being independently written studies on different target variables.

Reliability of the instrument itself is treated as something to demonstrate, not assume. A second coder, working directly from the raw papers and the PAVE-ML rubric and blind to the codes already assigned, independently coded a pilot sample of the corpus in randomised order, with the selection and blinding protocol documented in full alongside the coding materials. The protocol specifies Cohen's κ per field over a 15% double-coded sample, with any field falling below $\kappa = 0.60$ triggering a rewritten decision rule and a full recoding rather than a footnoted caveat; reconciling the pilot against the existing codes and extending it to that full sample is this review's single highest-priority remaining methodological task, named as such in §13, and the coding to date should be read accordingly as single-reviewer with an independent check under way rather than as already inter-rater-validated.

13. Research agenda

The corpus and the taxonomy built from it are sufficient to state a research agenda now, ahead of the wider audit a larger corpus would support. Ten directions follow, each stated as a testable question paired with the data and method that would settle it, rather than as a general aspiration.

1. **The missing baseline, at scale.** For every H1/H2 study reporting a hybrid model, does the paper also report the same base learner tuned conventionally, on the same partition and metric, under a comparable search budget? §9's evidence, drawn from the six within-paper comparisons close enough to test this directly, suggests the answer is usually no; establishing how often across the full corpus rather than this six-study sample is the natural next step.
2. **Search-budget parity.** Among H1/H2 studies, what fraction report the number of function evaluations, iterations, or wall-clock time spent tuning the hybrid arm versus the baseline arm? PAVE-ML item 12b is designed to make this answerable at scale.
3. **Optimiser proliferation versus optimiser value.** Dozens of named metaheuristics recur across this literature (§4). Does a newly proposed optimiser outperform two or more established optimisers on the same pavement problem, or only outperform an unspecified or untuned baseline? PAVE-ML item 12d targets this directly.
4. **Section-grouped partitioning for the CRCP substrate series.** At least nine papers by one overlapping author team share a single ~33-section CRCP LTPP substrate (§7), two of them full-text-confirmed to use an ungrouped split. A systematic re-derivation, from the LTPP database directly, of whether each remaining paper's published cross-validation folds respect section-level grouping would settle a question the other seven papers' abstracts do not answer, and is the single highest-value full-text verification task given how much of the corpus's rutting/cracking/IRI evidence traces to this one substrate.
5. **The H6 leakage question.** A wider harvest pass found H6 is rare but not absent (§8) — [48] is an abstract-confirmed wavelet-decomposition exemplar, and [49] a provisional EEMD-based one. Both were located too late for full-text access before this draft, and both publishers blocked automated fetching. Whether the decomposition step in either was fitted before or after the train/test split — the specific failure mode this taxonomy names for H6 — is the highest-priority open full-text verification task for the type, now that "does an H6 exemplar exist" is no longer the open question.

6. **Physics-informed learning has moved past one exemplar.** [21] was, in an earlier phase of this review, the seed corpus's only clear H3 case; a single 2026 harvest pass located nine further PINN studies published between 2024 and 2026 [62, 63, 64, 65, 66, 67, 68, 69, 70], none yet full-text verified. That a single targeted search surfaced this many suggests physics-informed learning in pavement engineering is past the single-exemplar stage and entering a genuine growth phase circa 2024–2026 — itself a bibliometric finding worth stating plainly rather than the more cautious one-exemplar framing this item originally posed as an open question. What remains genuinely open is comparative: does this cohort's physics constraint measurably outperform an unconstrained network on the same data, or is "physics-informed" becoming this literature's next under-audited novelty label the way "hybrid" already is (§1)?
7. **Interpretability on validated models only.** §10 and §11 argue that a SHAP or Garson decomposition computed on a model with no external validation ranks the structure of the training data rather than the physics of the pavement. What share of the corpus's interpretability-method studies pair the explanation with a validated model, by the external-validation criterion in §2.3?
8. **The efficiency dimension of hybridisation.** Most studies in this corpus report only accuracy gains. [40] is the seed corpus's only example reporting a feature-set reduction alongside an accuracy claim. A systematic accounting of computational-cost reporting (PAVE-ML item 19) would establish whether hybridisation's practical case rests on accuracy, efficiency, or an unexamined mixture of both.
9. **Deployment evidence as an outcome variable.** §9's deployment-evidence ladder (none through field trial) is, on current evidence, heavily weighted toward its lower rungs. Does deployment evidence correlate with any of the rigour-audit fields — external validation, baseline strength — or are the two independent, which would itself be a notable finding about how this field's incentives currently operate?
10. **Replication of the premium construct outside pavement engineering.** The tuned-baseline problem this review documents is unlikely to be unique to pavement engineering. A parallel audit in an adjacent data-driven civil engineering literature — concrete mix design, geotechnical site characterisation — would establish whether the pattern is domain-specific or a broader feature of how metaheuristic-learner hybrids are currently reported across civil engineering.

14. Conclusions

The four conclusions below are drawn from the 143 primary studies coded to date, of which 103 meet the structural definition of hybridity in §2.3. They are stated with the confidence this evidence base supports, and each is written so that the growing corpus can confirm, sharpen, or revise it without the underlying argument needing to be restructured.

First, a literature search restricted to the vocabulary of "hybrid" models undercounts this field's actual hybridisation activity, because a meaningful share of qualifying studies describe themselves through optimiser names, not through the word hybrid itself. The algorithm-driven search protocol in §2 was built as a remedy, and the 2026 harvest quantifies the gap it was designed to close directly: of 2,332 unique, pavement-gated records the algorithmic search returned, only 4.5% contain the word "hybrid" anywhere in title or abstract. A label-only search over the same candidate pool would therefore have missed the large majority of records an algorithm-name search retrieved — not a marginal undercount, but the dominant mode of how this literature is actually written.

Second, the taxonomy in §4 shows that "hybrid" currently functions as a single label over at least seven structurally and theoretically distinct couplings, each with its own characteristic failure mode. Collapsing them, as the field's current vocabulary does, obscures exactly the distinctions — between a metaheuristic searching hyperparameters and one searching weights, between architecture fusion and heterogeneous stacking — that determine whether a given failure mode is even in principle detectable from what a paper reports.

Third, and most consequentially, the comparison required to state a hybridisation premium — the same base learner, tuned conventionally, on the same data, under a comparable budget — is rarely made available by the studies that would need to report it. Of the studies examined closely enough in the seed corpus to test this directly, the missing baseline, not an unfavourable premium, is the dominant pattern (§9). Where the comparison approaches fairness, as in [1], the hybrid or deep-architecture advantage narrows or disappears against a properly tuned conventional alternative.

Fourth, the field has the practical means to fix this at essentially no added cost. None of the four hybrid-specific PAVE-ML items (§12) requires new data collection or a new experiment — reporting the base learner's own tuned performance, the search budget for each arm, the spread across repeated stochastic runs, and a comparison against established optimisers are all computable from runs a study has, in most cases, already performed. The gap this review documents is therefore substantially a reporting gap rather than a capability gap, which is the most optimistic reading available of an otherwise uncomfortable set of findings — and the reading this review's own instrument, PAVE-ML, is built to keep testing as the corpus underlying it continues to grow.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Declaration of generative AI in the writing process

During the preparation of this work, the authors used Claude (Anthropic) to assist with literature identification and screening (algorithmic query construction against the eligibility criteria in §2.3, structural — not lexical — candidate screening), construction and verification of the review database (every included record's DOI independently verified against the publisher record), and drafting of manuscript text. After using this tool, the authors reviewed and edited the output as needed and take full responsibility for the content of this publication.

Data availability

The screening database, coding rubric, harvesting scripts and the full analysis pipeline that produced every figure and table in this paper are available at https://github.com/sabernaseralavi-60/Does-Hybridisation-Pay-A-Systematic-Review-of-Hybrids-in-Pavement-Engineering.

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